

Key Terminology

Wafer

- A thin, round slice of silicon
- Used as the base to build computer chips
- One wafer holds many chips

Die (plural: dies)

- A single chip on a wafer
- Each die becomes one usable chip if it passes testing

Die-Level Testing (Wafer Probe Test)

- Electrical testing done before chips are separated
- Checks if each die works correctly

Yield

- The percentage of dies that work correctly
- Higher yield means lower manufacturing cost

Wafer Map

- A 2D picture of a wafer
- Shows which dies passed or failed testing

Pixel (in Wafer Maps)

- Represents one die
- Stores test results, not colors or brightness

Defect Pattern

- A visible shape formed by failing dies
- Often points to a manufacturing problem
- Examples: center failures, edge rings, scratches

Process Sensors

- Devices inside manufacturing machines
- Measure conditions like temperature, pressure, and power

Process Run / Production Instance

- A single execution of a manufacturing process, often corresponding to the processing of one wafer or lot under a specific set of tool conditions.

Introduction

- **Wafer map data**, which provides a spatial view of defects across a wafer after fabrication and testing
- **Process sensor data**, which captures physical and electrical conditions during wafer fabrication

While both data sources relate to the same manufacturing process, they capture fundamentally different aspects of it. Wafer maps describe *where* defects occur, whereas sensor data helps explain *why* those defects may have occurred.

Die-Level Electrical Testing

Before a wafer is cut into individual chips, it undergoes **die-level electrical testing**, commonly referred to as **wafer probe testing**.

During this stage:

- A probe card makes physical contact with each die on the wafer
- Electrical characteristics such as continuity, leakage current, and voltage thresholds are measured
- Each die is classified as passing, failing, or untested

The results of this testing stage provide the raw information used to construct wafer maps.

What Is a Wafer Map?

A **wafer map** is a two-dimensional grid representation that preserves the physical layout of dies on a wafer.

- Each grid cell corresponds to a single die
- The spatial arrangement reflects the physical location of dies on the wafer
- The circular shape of the wafer naturally emerges from the grid structure

How Wafer Map Pixels Are Generated

In wafer maps, a **pixel does not represent light intensity**, as in conventional digital images.

Instead:

- Each pixel encodes the **electrical test outcome of a single die**
- Pixel values are discrete and symbolic

Typical encoding:

- `0` → No die or untested location
- `1` → Die passed electrical test
- `2` → Die failed electrical test

Thus, each pixel corresponds directly to a physical die on the wafer.

Spatial Defect Patterns

Because dies are arranged spatially across the wafer surface, defects often form recognizable geometric patterns:

- **Center defects** — commonly associated with lithography focus or exposure issues
- **Edge-ring defects** — linked to temperature gradients or etching non-uniformity
- **Scratch defects** — caused by mechanical handling or tool contact
- **Random defects** — typically resulting from particle contamination

These spatial regularities make wafer maps well suited for image-based machine learning models such as convolutional neural networks (CNNs).

Process Sensors in Semiconductor Manufacturing

Unlike wafer maps, which are generated after fabrication, **process sensors operate during fabrication**.

Sensors are embedded within manufacturing tools such as etchers, deposition chambers, and lithography systems, and they monitor:

- Temperature
- Pressure and vacuum levels
- Gas flow rates
- Plasma density
- Radio-frequency (RF) power
- Chemical concentrations
- Tool health and calibration indicators

These sensors provide continuous insight into the conditions under which wafers are manufactured.

Example: Wafer Map Insight (Where Defects Occur)

What we observe

- Many failed dies form a ring near the wafer edge
- The pattern is consistent across multiple wafers

What this tells us

- The problem is spatial, not random
- Defects are linked to wafer position

Conclusion

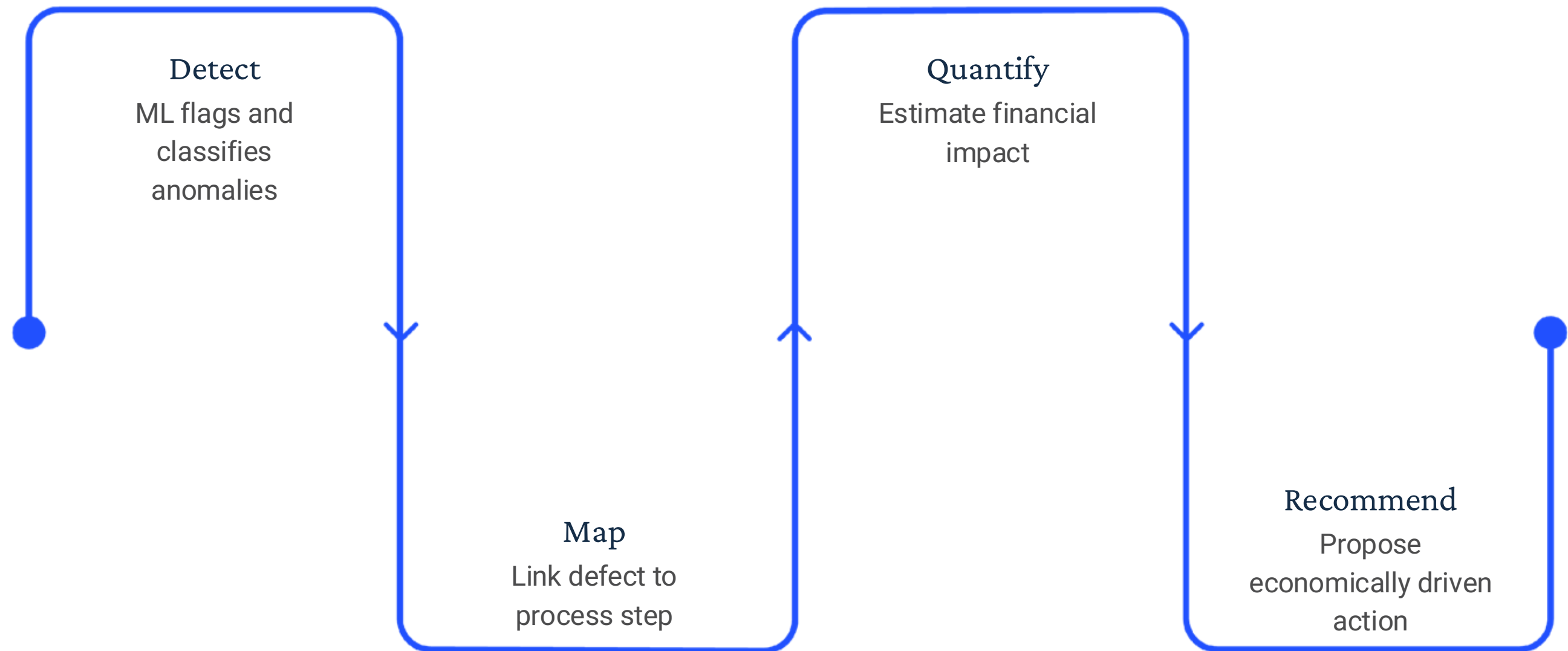
- The issue likely comes from a process that affects wafer edges
- Examples: uneven heating or etching near the boundary

References

- Kyeong, S., Kim, H., & Kim, S. (2020). Classification of wafer map defects based on deep learning methods with a small amount of data. IEEE Access, 8, 21796–21806.
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- Li, Y., Zhang, X., & Wang, J. (2024). Deep learning approaches for wafer defect pattern recognition. arXiv Preprint. <https://arxiv.org/html/2411.11029v1>
- JMP Community. (2021). How to make a wafer map in JMP in under 30 seconds.
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- MathWorks. (n.d.). Classify defects on wafer maps using deep learning.
<https://www.mathworks.com/help/vision/ug/classify-defects-on-wafer-maps-using-deep-learning.html>
- PDF.com. (n.d.). Understanding semiconductor data visualization with wafer maps: An introduction. <https://www.pdf.com/understanding-semiconductor-data-visualization-with-wafer-maps-an-introduction/>

- Wikipedia contributors. (n.d.). Substrate mapping.
https://en.wikipedia.org/wiki/Substrate_mapping

Strategic Shift: From Anomaly Detection to Decision Intelligence



Input: Wafer map anomalies report from production batch

Output: Prioritized repair recommendations for specific tools

Example Mapping: The "Donut" Defect Case Study

Defect: Donut

Likely Causes

- CMP pad wear (primary)
- Lithography redeposition (secondary)

Financial Impact

- Yield Impact: 30%
- Repair Cost: \$150,000
- Replacement: \$5.25M
- Downtime: \$1M/hr
- Frequency: Medium occurrence (18% of wafer defects)

[Source:](#) Semiconductor Digest (*A dirty Job: The Challenge of CMP*), EE Time (*CMP foundry expands services, capacity*)

OUTPUT

Example Executive Output: Actionable Decision Card

Defect: Edge_Ring Likely Process: Etch (Confidence: High)	Estimated Daily Loss: \$180k 30-Day Projection: \$5.4M Cost Basis: Tool-level throughput loss	Repair Cost: \$120k Replacement Cost: \$12M Frequency of Repair: Low Repair Lifetime: 12 months	ROI if Repaired: 3500% Recommended Action: Replace focus ring + recalibrate RF (Confidence: High)
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Source: Knowledge Sourcing Intelligence (*Semiconductor Dry Etch Systems Market - Strategic Insights and Forecast (2025-2030)*), Science Direct (*Edge Exclusion*), Semi (*When Minutes Matter: Time-Critical Logistics as a Competitive Edge*)

WaferGuard ML

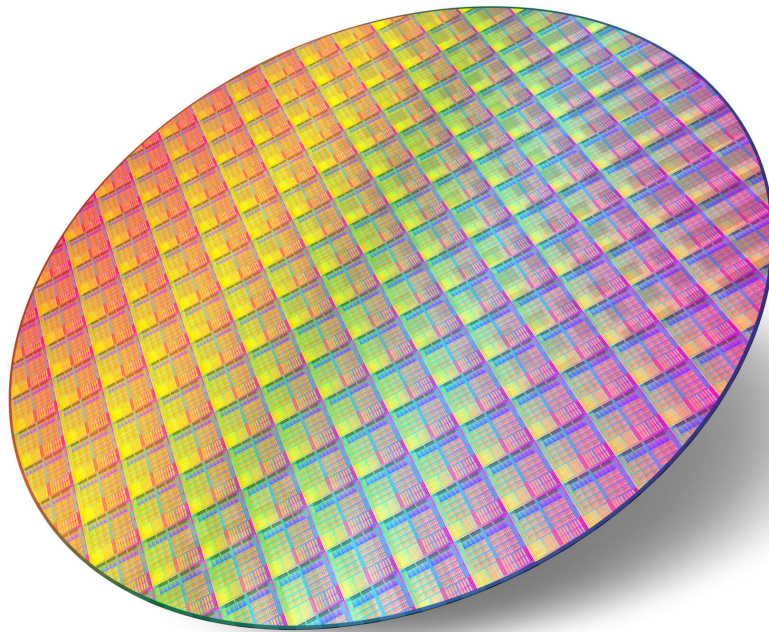
Sprint 1: WM811K Dataset Exploration

Deloitte AI Team

2026-02-01

What is a Semiconductor Wafer?

A semiconductor wafer is a thin, circular slice of silicon used to manufacture computer chips.



- **Thin circular disk** (like a CD, but 12 inches wide)
- **Pure silicon crystal** grown in labs
- **Hundreds** of transistors per mm^2

Silicon wafer = computer chip foundation

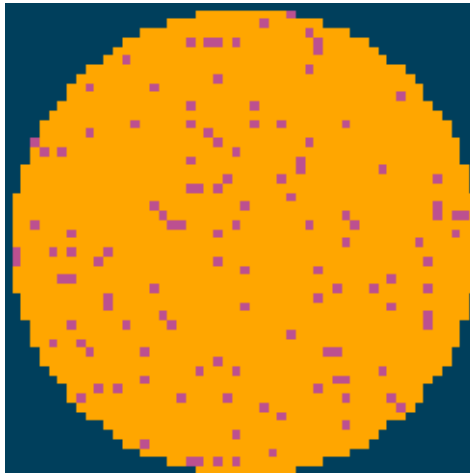
The Challenge

- Traditional manual inspection is slow and inconsistent
- Defects detected too late in production
- Lost productivity = Millions in yield loss
- Need real-time, automated detection

Wafer Images Dataset

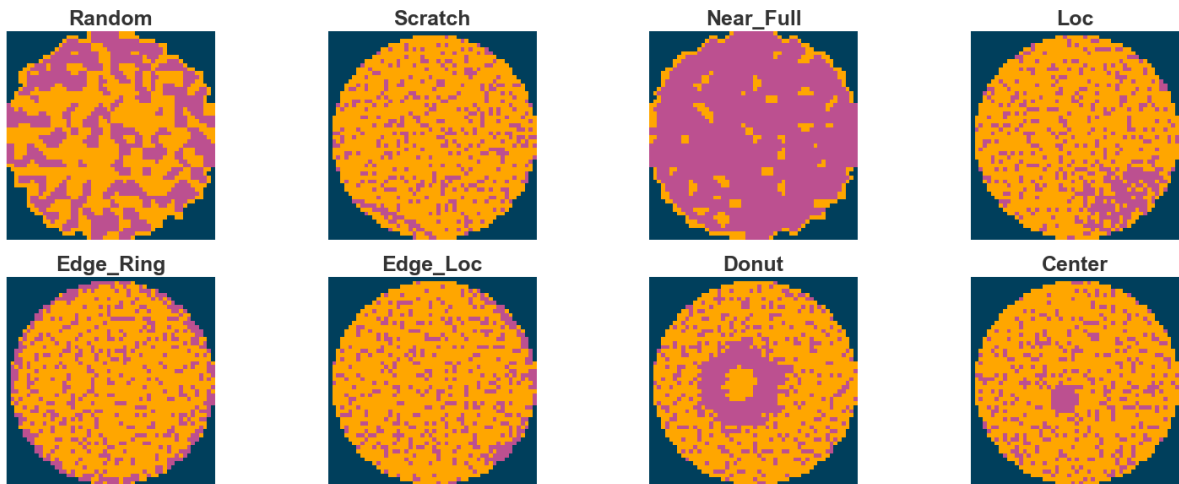
- The data shape is 52×52 .
- 0 means blank spot
- 1 represents normal die that passed the electrical test
- 2 represents broken die that failed the electrical test.
- These wafer maps are obtained by testing the electrical performance of each die on the wafer through test probes.

Example of a Normal Wafer

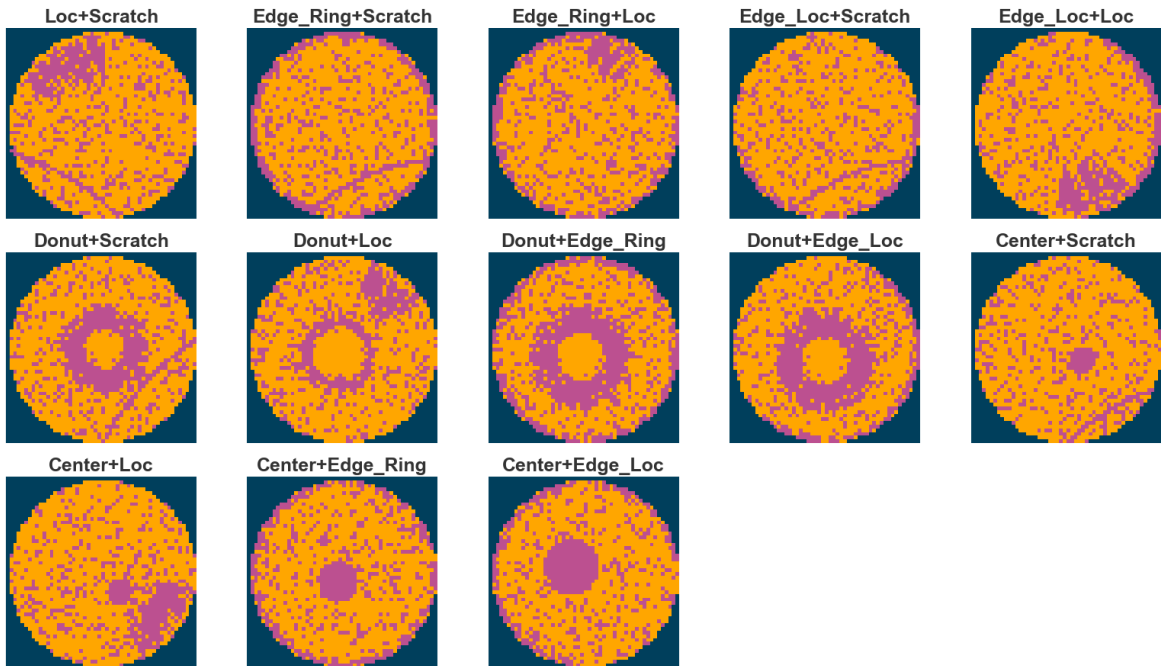


Defect Patterns

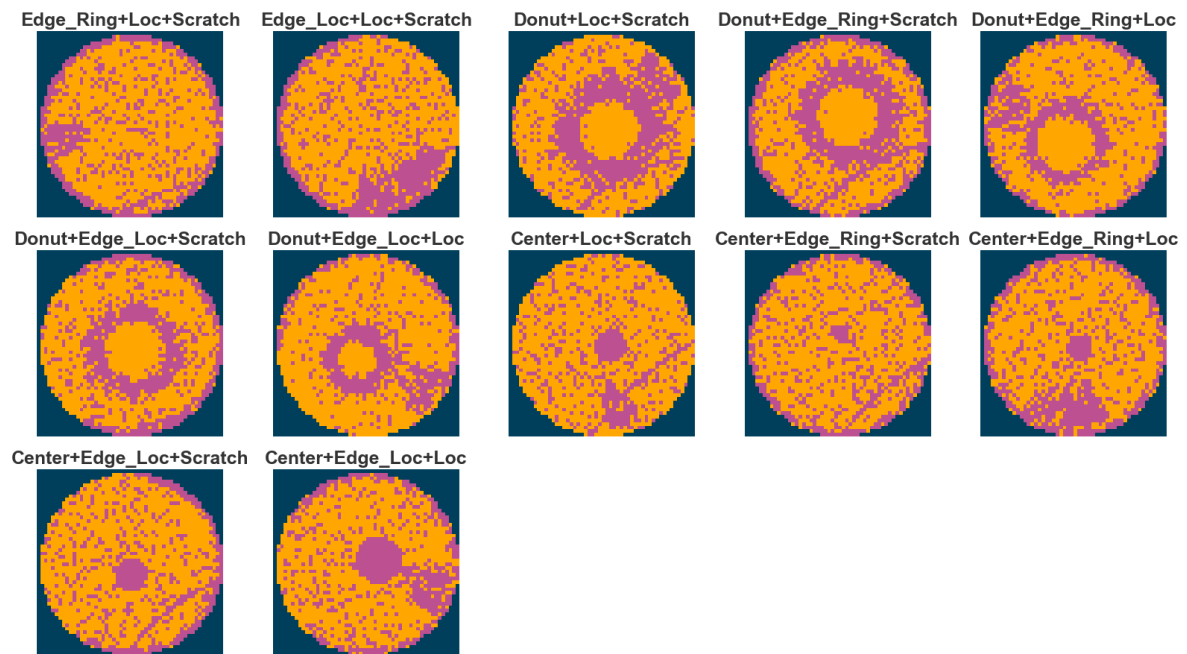
8 distinct defect categories analyzed in our dataset



Two Types of Multi-Defect Patterns

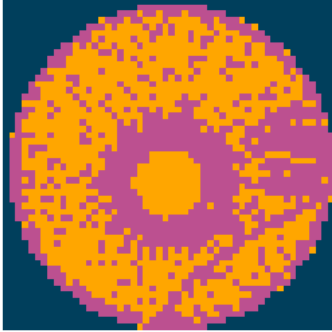


Three Types of Multi-Defect Patterns

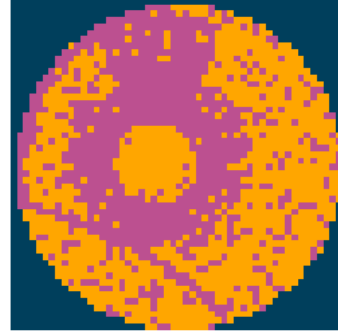


Four Types of Multi-Defect Patterns

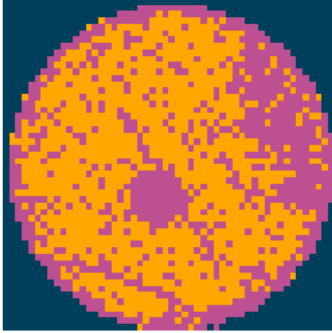
Donut+Edge_Ring+Loc+Scratch



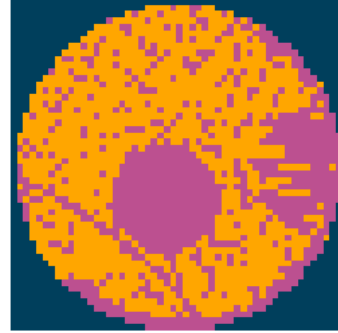
Donut+Edge_Loc+Loc+Scratch



Center+Edge_Ring+Loc+Scratch



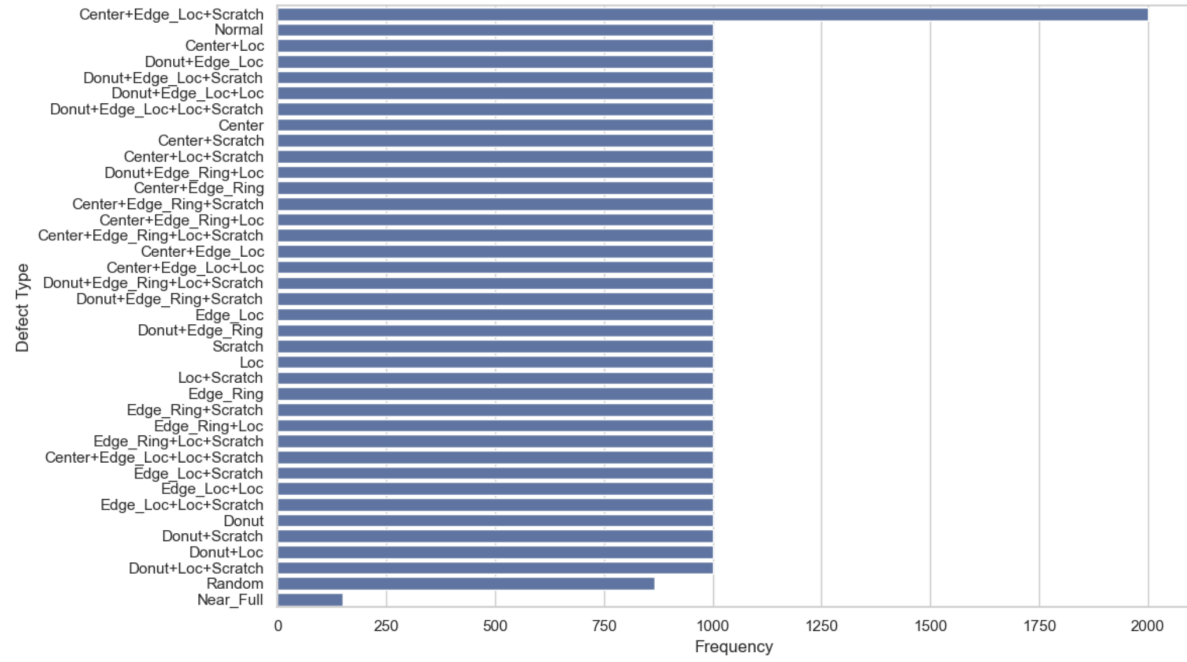
Center+Edge_Loc+Loc+Scratch



Defect Pattern Labels

- The labels in the data are using one-hot encoding, a total of 8 dimensions, corresponding to the 8 basic types of wafer map defects (single defect).
 - Normal: [0, 0, 0, 0, 0, 0, 0, 0]
 - Center: [1, 0, 0, 0, 0, 0, 0, 0]
 - Donut: [0, 1, 0, 0, 0, 0, 0, 0]
 - Edge-Loc: [0, 0, 1, 0, 0, 0, 0, 0]
 - Center-Donut-Edge-Loc: [1, 1, 1, 0, 0, 0, 0, 0]

Pattern Distribution



Dataset Summary

- 38K Wafer Maps
- 52x52 Pixel Resolution
- One-Hot Encoded Pattern Labels
- 38 types in the mixed-type wafer map defect dataset
 - 1 normal type
 - 8 single defect types
 - 29 mixed-type (2: 13, 3: 12, 4: 4) defect types.

Data Quality: Excellent

- No missing values
- Complete Coverage of Defect Types
- Mostly Balanced Classes
- Ready for Modeling

WaferGuard ML

Sprint 1: SECOM Dataset Exploration

Deloitte AI Team

2026-02-01

What is SECOM Data?

Sensor data from a semiconductor manufacturing process.

- **590 sensor measurements** per production entity
- **1,567 samples** (wafers/chips)
- **Binary outcome:** Pass (-1) vs Fail (1)
- **Source:** UCI Machine Learning Repository

Real-world sensor data is notoriously “messy” and redundant

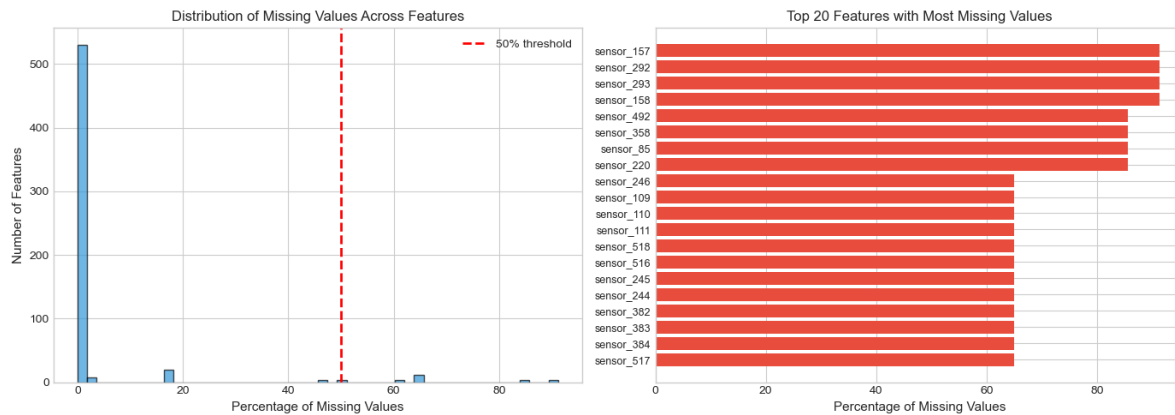
The Challenge

- High-dimensional sensor data (590 features)
- Severe class imbalance (~93% Pass, ~7% Fail)
- Significant missing values across features
- Many redundant/correlated sensors

Dataset Overview

Attribute	Value
Samples	1,567 production entities
Features	590 sensor measurements
Labels	Pass (-1) / Fail (1)
Collection Period	89 days (Jul-Oct 2008)

Missing Value Analysis

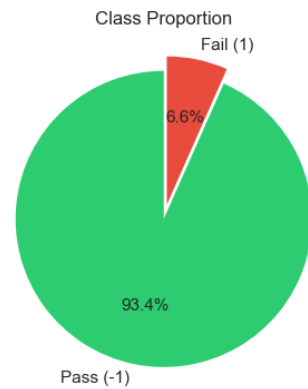
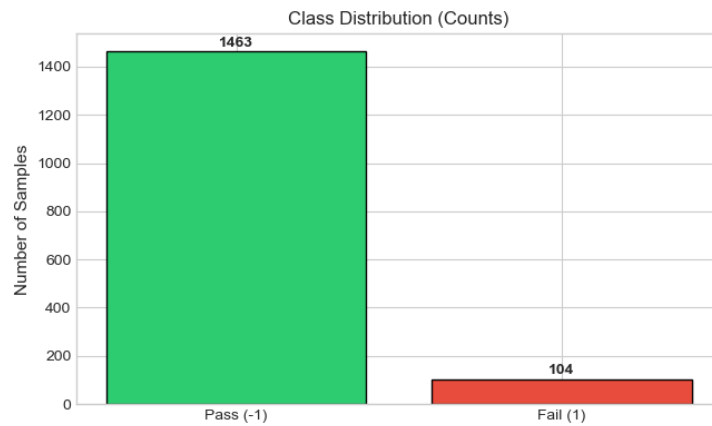


Missing Value Distribution

- **Features with ANY missing:** 538 out of 590 (91%)
- **Features with >50% missing:** 28 features
- **Average missing per sample:** 26.8 features (4.5%)

Strategy: Drop features with >50% missing; impute the rest

Descriptive Statistics

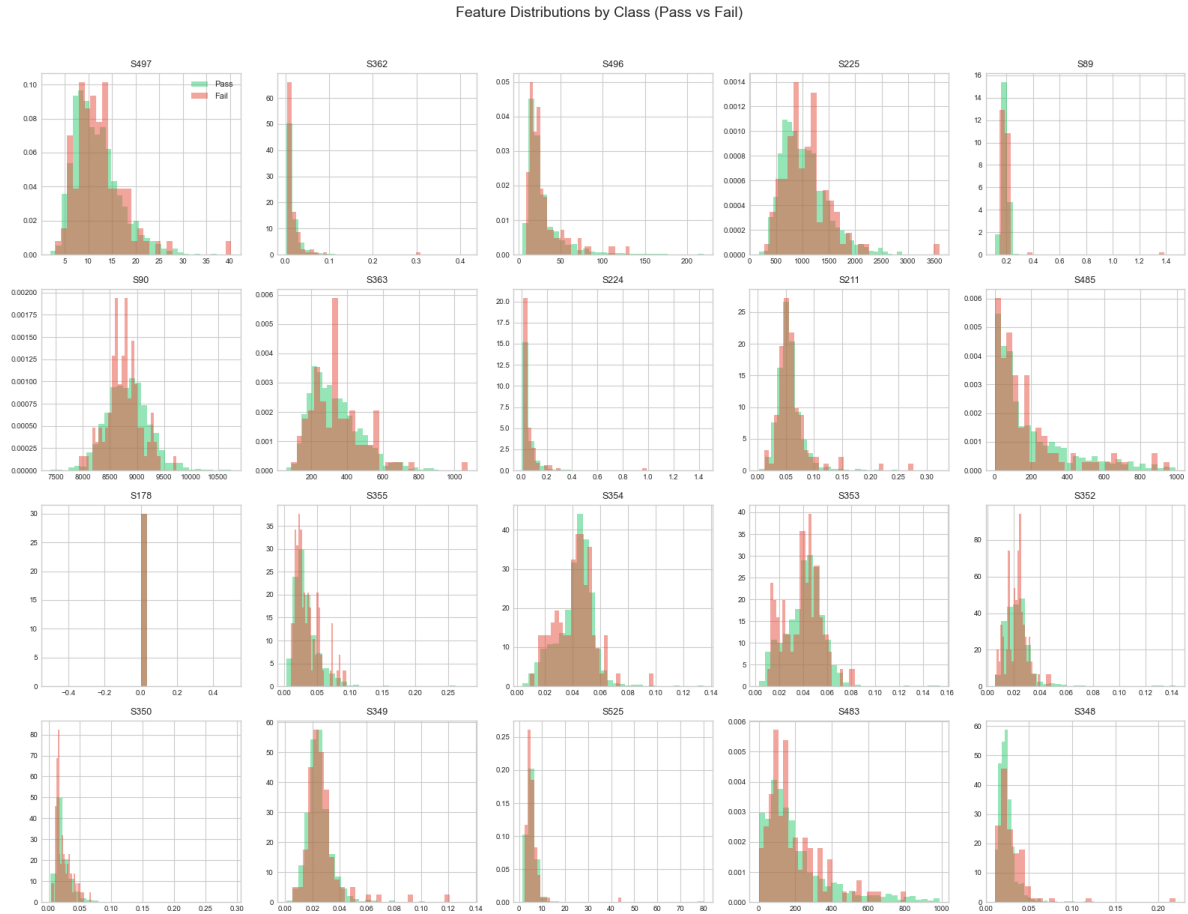


Constant Features

Some sensors show **zero variance** (useless for prediction).

- Example: `sensor_5 = 100.0` for all samples
- These provide no discriminative power
- Must be identified and removed

Class Distribution



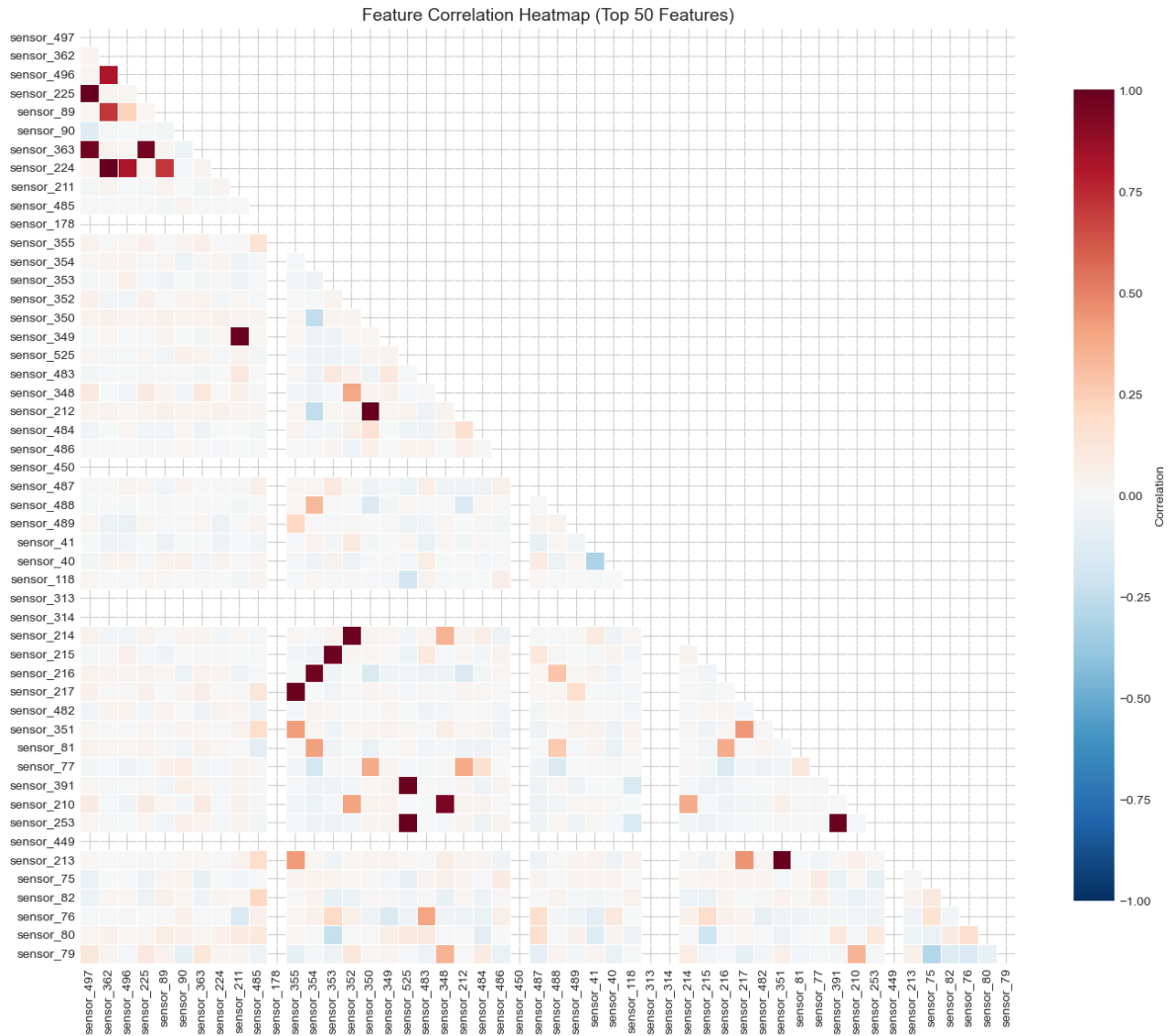
Class Imbalance

The dataset is heavily skewed towards “Pass”.

Class	Label	Frequency
Pass	-1	~93%
Fail	1	~7%

Risk: A model predicting “Pass” everywhere achieves 93% accuracy but fails the objective.

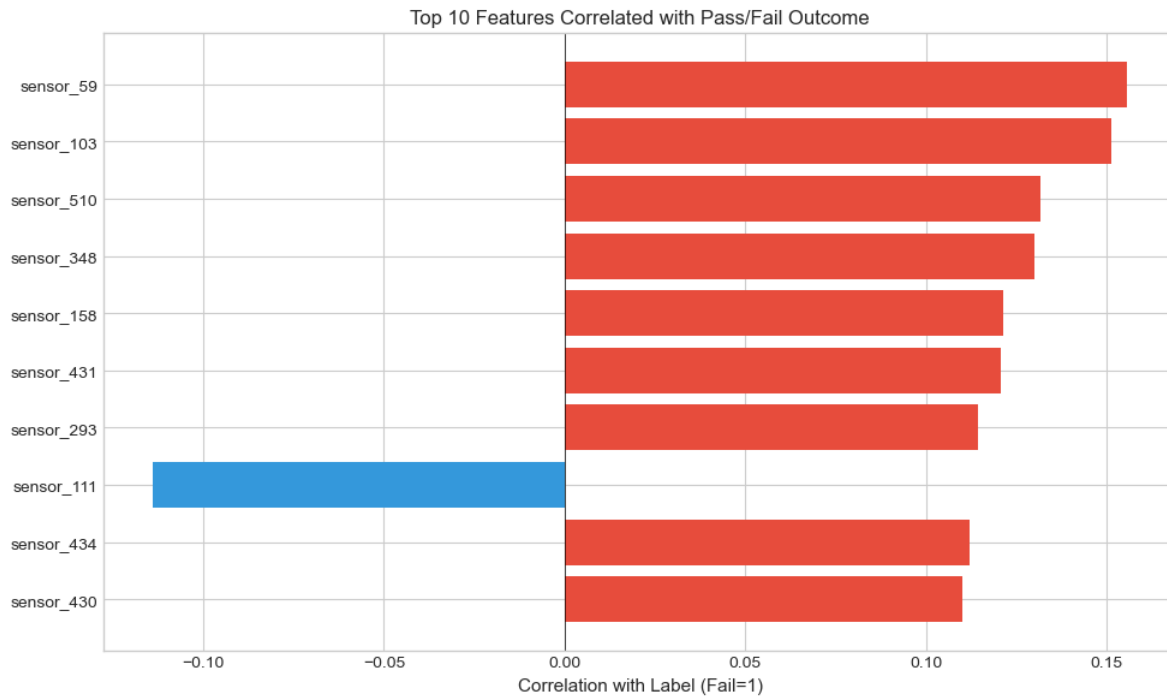
Correlation Analysis



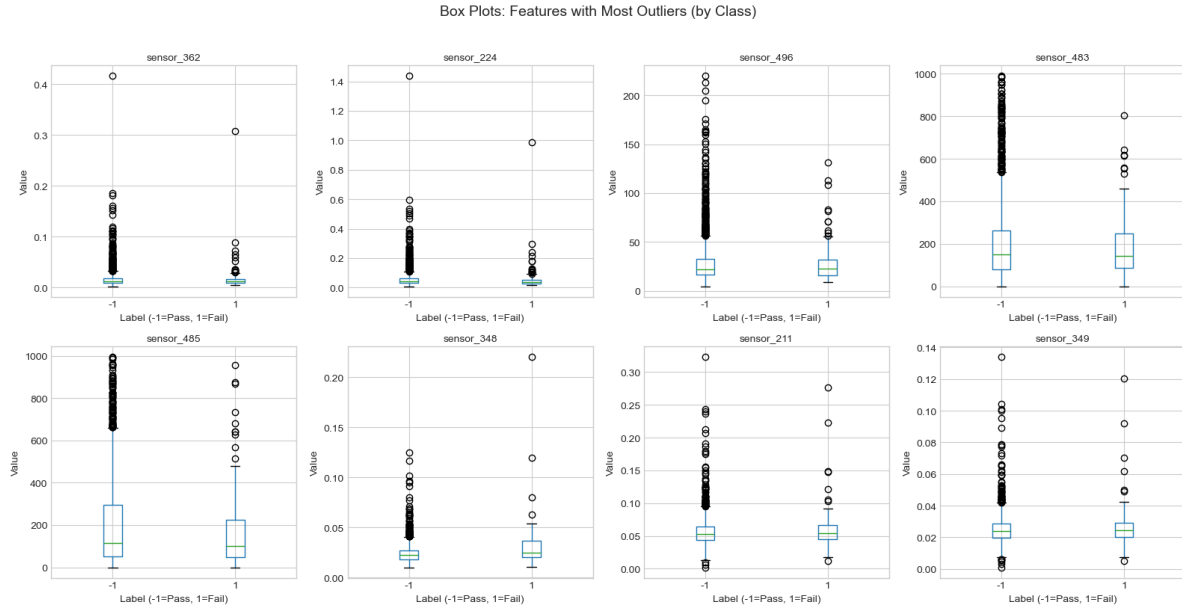
Feature Correlations

- Many sensors are highly correlated
- Redundant features increase model complexity
- Strategy: Remove one of each highly correlated pair ($|r| > 0.95$)

Additional Analysis



Feature Distribution



Preprocessing Strategy

1. Cleaning:

- Remove constant features (Zero Variance)
- Remove high missingness columns ($>50\%$)

2. Feature Reduction:

- Address multicollinearity ($|r| > 0.95$)
- Consider PCA

Handling Imbalance

To enable effective defect detection:

- **Stratified Sampling:** Maintain 7% failure rate in splits
- **SMOTE:** Generate synthetic failure examples
- **Class Weights:** Penalize missing a failure more heavily

Next Steps

1. Execute preprocessing pipeline
2. Baseline Model Training (Logistic Regression / Random Forest)
3. Evaluate using **Recall** and **F1-Score** (Accuracy is misleading here)

Dataset Summary

- 1,567 Production Samples
- 590 Sensor Features
- Binary Pass/Fail Labels
- 89 Days of Production Data

Data Quality Assessment

- Significant Missing Values (91% of features affected)
- Constant Features Present
- Severe Class Imbalance (93/7 split)
- Preprocessing Pipeline Defined
- Ready for Model Development